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ARTICLE

First-in-human study evaluating safety, pharmacokinetics, and pharmacodynamics of lorundrostat, a novel and highly selective aldosterone synthase inhibitor

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Abstract

Dysregulation of the mineralocorticoid hormone aldosterone is an increasingly prevalent cause of hypertension. Aldosterone synthase (CYP11B2) shares 93% homology to 11β-hydroxylase (CYP11B1), which produces cortisol. Lorundrostat, a highly selective inhibitor of CYP11B2, is a potential safe and effective treatment for aldosterone-dependent, uncontrolled hypertension, including treatment-resistant hypertension. Lorundrostat showed highly selective inhibition of CYP11B2 in vitro, with 374-fold selectivity for CYP11B2 vs. CYP11B1. A first-in-human study of single ascending doses ranging from 5 to 800 mg and multiple ascending doses ranging from 40 to 360 mg once daily was conducted in healthy participants. After single- and multiple-dose administration, lorundrostat plasma levels peaked 1–3 h after administration with a $t_{1/2}$ of 10–12 h. Plasma aldosterone decreased up to 40% with single 100-mg to 200-mg doses and up to 70% with single 400 to 800-mg doses. Plasma aldosterone returned to baseline within 16 h after single 100-mg doses and multiple once-daily 120-mg doses. Lorundrostat demonstrated a favorable safety profile in healthy participants. Dose- and exposure-dependent inhibition of renal

Abbreviations: 11-DOC, 11-deoxycorticosterone; ACE, angiotensin-converting enzyme; ACTH, adrenocorticotropic hormone; AE, adverse event; AUC, area under the curve; $AUC_{0-\infty}$, area under the curve extrapolated to infinity; AUC_{0-last}, area under the curve from 0 to last measurable concentration; $AUC_{0-\tau}$, area under the curve from 0 to dosing interval; C_{max}, maximum concentration; CYP, cytochrome P450; GPER, G protein-coupled estrogen receptor 1; GPR30, G protein-coupled receptor 30; hCYP, human cytochrome P450; K2-EDTA, dipotassium ethylenediaminetetraacetic acid; MAD, multiple ascending dose; MR, mineralocorticoid receptor; MRA, mineralocorticoid receptor antagonist; PAC, plasma aldosterone concentration; PCC, plasma cortisol concentration; PD, pharmacodynamic; PK, pharmacokinetic; RAAS, renin–angiotensin–aldosterone system; SAD, single ascending dose; $t_{1/2}$, half-life; TEAE, treatment-emergent adverse event; T_{max}, time to reach the maximum concentration.

Hidetoshi Shimizu and Michael A. Tortorici are co-first authors with equal contribution.

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tubular sodium reabsorption was observed across a clinically relevant dose range with no suppression of basal or cosyntropin-stimulated cortisol production and only a modest increase in mean serum potassium.

Study Highlights

WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

Applying current (2017) hypertension guidelines of controlled blood pressure < 130/80 mmHg to NHANES 2017–2020 data, 77.5% of US adults with hypertension had uncontrolled blood pressure. Aldosterone plays an important role in uncontrolled hypertension, including treatment-resistant hypertension. Aldosterone synthesis inhibition is a viable treatment approach. The development of selective inhibitors of CYP11B2 (aldosterone synthase) suitable for human use has proven to be challenging.

WHAT QUESTION DID THIS STUDY ADDRESS?

This study aimed to characterize the safety, tolerability, pharmacokinetics, and pharmacodynamics of single and multiple oral doses of lorundrostat in healthy adult participants.

WHAT DOES THIS STUDY ADD TO OUR CURRENT KNOWLEDGE?

A highly selective CYP11B2 inhibitor, lorundrostat, demonstrated a favorable safety profile in healthy human participants. Lorundrostat also demonstrated optimized pharmacokinetics and evidence of inhibition of renal tubular aldosterone signaling across a broad dose range with no suppression of basal or stimulated cortisol production. Lorundrostat shows promise for patients with uncontrolled hypertension, including those with treatment-resistant hypertension, in which renin–angiotensin–aldosterone-system inhibition (RAASi) does not lead to adequate suppression of aldosterone production. In contrast to mineralocorticoid receptor antagonists (MRAs), lorundrostat is designed to reduce rather than stimulating the substantial increase in circulating aldosterone seen with MRAs. This unique difference is anticipated to reduce adverse non-MR-mediated aldosterone signaling via the GPER (GPR30) receptor on blood vessels and other tissues, avoiding the estrogenic side effects of MRAs and preventing "aldosterone breakthrough" during RAASi treatment.

HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY OR TRANSLATIONAL SCIENCE?

These results demonstrate that lorundrostat has a favorable safety, pharmacokinetic, and pharmacodynamic profile and will inform dosing in later stage clinical studies in patients with uncontrolled hypertension, including those with treatment-resistant hypertension.

INTRODUCTION

Lorundrostat is a novel molecular entity (Figure 1) designed to target the renin–angiotensin–aldosterone system (RAAS), via inhibition of aldosterone synthase (CYP11B2) resulting in targeted suppression of aldosterone production. The RAAS plays a central role in the control of intravascular volume, blood pressure, and serum potassium concentration.[1,2] Normally, short and long negative feedback loops maintain homeostasis in the RAAS pathway.[3] Low intravascular volume leads to increased renin- and angiotensin-converting enzyme (ACE)-mediated production of angiotensin-2 and subsequent production of aldosterone.[2] Increased aldosterone ultimately increases sodium reabsorption in the distal nephron, leading to increased intravascular volume, reducing pro-renin production, closing the long negative feedback loop, and maintaining homeostasis.[1]

Additionally, aldosterone binds the mineralocorticoid receptor (MR) in varied tissues, including brain, heart, vasculature, and adipose tissue, resulting in other diverse biological activities, such as visceral adipokine

release, inflammation and other systemic factors contributing to the development and progression of cardiovascular and renal disease.[1,4,5] These effects are likely responsible for the findings that patients with excess aldosterone suffer more cardiovascular incidences than essential hypertension patients with the same level of blood pressure.[6,7]

Because of the importance of the RAAS pathway in maintaining control of volume and blood pressure, each step in the pathway has been explored as a therapeutic target for the treatment of hypertension. The earliest pathway inhibitor used in humans was spironolactone, a steroidal inhibitor of the MR.[5,8] As newer agents have been developed, use of MR antagonists (MRA) in the treatment of hypertension has largely been supplanted by use of upstream pathway inhibitors, including ACE inhibitors (ACEi), angiotensin receptor blockers (ARBs), and renin inhibitors. While these agents have benefits in reducing blood pressure, as well as morbidity and mortality due to cardiovascular and renal disease, there are several mechanistic and off-target adverse effects that can complicate the use of RAAS pathway inhibitors. A study in left ventricular dysfunction (RESOLVD) showed that RAAS suppression with an ACEi and an ARB did not suppress long-term aldosterone secretion,[9] a phenomenon termed "aldosterone breakthrough." Complicating the situation further is the increasing recognition of dysregulated aldosterone and obesity-related dysregulated aldosterone production via visceral adipokines as underlying causes of treatment-resistant hypertension.[5,10,11]

Overall, the global prevalence of treatment-resistant hypertension is estimated to be 10.3%, but the rate is higher among patients with chronic kidney disease, renal transplant recipients, and older adults (22.9%, 56.0%, and 12.3%, respectively).[12] Given the potential limitations of upstream inhibition of the RAAS with ACEi or ARBs, including aldosterone breakthrough, targeted inhibition of aldosterone synthesis may provide a more effective blockade of aldosterone-mediated cardiovascular diseases.

Aldosterone synthase (CYP11B2) is a mitochondrial cytochrome P450 (CYP) enzyme that converts 11-deoxycorticosterone (11-DOC) to aldosterone in three consecutive steps: 11-DOC converts to corticosterone, which converts to 18-hydroxycorticosterone, which converts to aldosterone.[1] CYP11B1, a key enzyme in glucocorticoid biosynthesis, has a high homology to CYP11B2 (>93%). High selectivity for CYP11B2 over CYP11B1 is an essential characteristic for a successful aldosterone synthase inhibitor.[13] Until recently, the development of improved and more selective inhibitors of CYP11B2 that are suitable for the treatment of hypertension has proven to be challenging. Osilodrostat, the first aldosterone synthase inhibitor, lacked high selectivity, resulted in blunted adrenocorticotropic hormone-stimulated cortisol release in some patients, and instead became a treatment for Cushing's disease.[14,15] The effects of lorundrostat on aldosterone suppression have been examined in preclinical pharmacology studies conducted in rats and monkeys (data on file). This first-in-human dose escalation study evaluates the safety, tolerability, pharmacokinetics (PK), and pharmacodynamics (PD) of single and multiple oral doses of lorundrostat, a highly selective CYP11B2 inhibitor. In addition, this study also compared lorundrostat PK in females and males and older adult (≥65 years old) and younger adult participants.

METHODS

In vitro studies

Assay of hCYP11B2 and hCYP11B1 inhibition

The inhibitory effects of lorundrostat on human CYP11B1 (hCYP11B1) and human CYP11B2 (hCYP11B2) enzyme activity were evaluated by determining the enzymatic conversion rates of 11-deoxycortisol to cortisol for hCYP11B1 and 11-DOC to aldosterone for hCYP11B2. The inhibition constant of lorundrostat (free base) on hCYP11B2 was calculated based on the rate of aldosterone generation from the substrate 11-DOC with the mitochondrial fraction of V79 cells stably expressing hCYP11B2. The inhibition constant of lorundrostat on hCYP11B1 was calculated based on the rate of cortisol generation from 11-deoxycortisol with the mitochondrial fraction of V79 cells stably expressing hCYP11B1. Osilodrostat, which is a potent inhibitor of both hCYP11B1 and hCYP11B2, was used as a positive control.

Phase I study

Study design

A four-part, randomized, double-blind, placebo-controlled, first-in-human study was performed (single-center in the Netherlands; November 16, 2016–May 01, 2017) to determine the safety, tolerability, PK, and PD of single ascending doses (SADs; part 1) and multiple ascending doses (MADs; part 2) of lorundrostat free base in a capsule in healthy participants (EudraCT Number: 2016–003500-32). The study also assessed sex- (part 3) and age-related (part 4) effects on the PK of a single dose of lorundrostat. An exploratory comparison of the effect of food on PK profiles of lorundrostat was also performed by assessing single-dose PK profiles of lorundrostat in the fasted state (part 1) vs.

PK profiles of lorundrostat in the fed state (part 2 after the first-dose administered). The study was approved by the local independent ethics committee; the BEBO foundation in Assen, Netherlands; and the Central Committee on Research Involving Human Subjects in The Hague, Netherlands; and conducted according to the provisions of the Declaration of Helsinki. Written informed consent was obtained from each study participant before conducting any protocol-related procedures.

Participants

Healthy male Caucasian participants (parts 1 and 2) aged 18–55 years (inclusive), healthy non-pregnant and non-lactating female Caucasian participants (part 3) aged 18–55 years (inclusive), and healthy male Caucasian participants aged ≥65 years (part 4) were eligible. Complete inclusion and exclusion criteria are given in Table S1.

Study stages

In part 1, in each dose cohort, six participants were randomized to receive a single dose of lorundrostat and two participants were randomized to receive matching placebo in the fasted state. All cohorts included two sentinel participants of whom one received lorundrostat and one received matching placebo. The remaining six participants, of whom five received lorundrostat and one received placebo, were dosed ≥24 h later. The dosing cohorts were 5, 10, 20, 50, 100, 200, 400, and 800 mg (5-mg starting dose rationale provided in Table S1). Doses were administered in an ascending order per cohort. After each cohort, all safety data were reviewed prior to initiating the next dose cohort.

In part 2, in each cohort, nine participants received daily doses of lorundrostat and three participants received matching placebo in the fed state from Day 1 to Day 7. Progression to the next dose level was based on available data from part 1 and the preceding dose cohort in part 2. The final dose levels of lorundrostat for part 2 were 40, 120, and 360 mg. An adrenocorticotropic hormone (ACTH) challenge test was performed on Day -2 and Day 6 to evaluate the lorundrostat selectivity for aldosterone synthesis.

Part 3 (sex-related effects) and part 4 (age-related effects) evaluated safety, tolerability, and PK of lorundrostat. In one cohort of eight females (part 3) and one cohort of eight males aged ≥65 years (part 4), six participants were randomized to receive a single 100-mg dose of lorundrostat and two participants were randomized to receive matching placebo in the fasted state in each part. The results were compared with those at the same dose in part 1.

Safety end points and assessments

Safety assessments included treatment-emergent adverse events (TEAEs), treatment-related TEAEs, vital signs, 12-lead electrocardiograms (ECGs), clinical laboratory evaluations, and physical examinations.

Statistical, pharmacokinetic, and pharmacodynamic analysis

Pharmacokinetics

The concentrations of lorundrostat in plasma (all parts of the study) and urine (part 2) were assessed by collecting plasma and urine at planned timepoints. Plasma PK parameters were derived by non-compartmental analysis using WinNonlin® (version 6.3) tool. Concentrations reported below the limit of quantification (BLQ) were set to 0. Area under the curve (AUC) was calculated using the linear trapezoidal method and using actual times. Dose proportionality for AUC extrapolated from 0 to infinity ($AUC_{0-\infty}$), from 0 to last measurable concentration (AUC_{0-last}), from 0 to 24 h (AUC_{0-24}, MAD only), and maximum concentration (C_{max}) was assessed using the power model. A linear model ($\ln(Y)=\alpha+\beta\times\ln(X)$, where Y is the pharmacokinetic parameter and X is the dose) was used to fit the power model, after log-transformation of the parameters. Dose proportionality was concluded if the 95% confidence interval (CI) for slope (β) included the value 1. Ratio of accumulation (MAD only) was calculated as AUC_{0-24} [Steady-State]/AUC_{0-24} [Day 1].

AUC_{0-last}, $AUC_{0-\infty}$, and C_{max} were used to explore any sex- and age-related effects (parts 3 and 4, respectively). A linear model was used to analyze log-transformed AUC and C_{max} with sex or age as fixed effect. Difference in least square means and corresponding 90% CI were back-transformed to obtain the estimates of geometric mean ratios and their CI for females vs. males and participants ≥65 years old vs. 18–55 years old. The result was judged to be statistically significant if the 90% CI did not include 1.

No formal statistical comparison was performed for an exploratory comparison of PK between fasted (part 1) and fed (part 2, after the first-dose administered) states.

Pharmacodynamics

Plasma concentrations of aldosterone, 11-deoxycortisol, and ACTH; renin activity; renin concentration; and serum concentrations of cortisol and 11-DOC were measured in all parts of the study. The amounts of aldosterone,

cortisol, sodium, and potassium excreted and the urinary $\log_{10}$ ($10\times Na^+/K^+$) ratio were measured per collection interval. Plasma concentrations of aldosterone and 11-deoxycortisol and serum concentrations of cortisol and 11-DOC were measured pre-ACTH dose and 30- and 60-minutes post-ACTH dose on Day -2 and Day 6.

PD parameters were derived by non-compartmental analysis using WinNonlin® Professional (version 6.3). AUC_{0-24} was determined where possible for plasma aldosterone concentration (PAC) and from 0 to 72 h (AUC_{0-72}) for serum cortisol concentration. The calculated AUC parameters were analyzed using a linear model. The log-transformed parameter of interest was the dependent variable with dose group (each lorundrostat dose and pooled placebo) as the fixed effect and log-transformed AUC_{0-24} on Day -1 as the covariate.

Log-transformed change from baseline (time-matched) from 0 to 24 h post-dose in plasma aldosterone and serum cortisol as repeated measures was analyzed using a linear mixed model with dose group (each active lorundrostat dose and pooled placebo) as fixed effect, corresponding time-matched values on Day -1 as covariate, with unstructured covariance. The pre-dose timepoint on Day 1 was used as the 24-h timepoint on Day -1.

Additional methods related to the in vitro studies and the validated quantitative measurement methods that were used for PK and PD markers are summarized in the Table S2.

RESULTS

In vitro hCYP11B2 and hCYP11B1 inhibition

Lorundrostat inhibited hCYP11B2 and hCYP11B1 with inhibition constant values of 1.27 nmol/L and 475 nmol/L, respectively. These results indicate that lorundrostat inhibits CYP11B2 with 374-fold selectivity over CYP11B1 (Figure 1).

Participant disposition and demographics

A total of 245 participants were screened, of which 116 were randomized. All participants randomized in parts 1, 3, and 4 completed the study as per protocol. One participant in part 2 did not complete the study. Participants' demographics are provided in Table 1. Part 1 (SAD) included 64 male participants between 18 and 55 years old. Part 2 (MAD) included 36 male participants between 19 and 54 years of age. Part 3 (sex effect) included eight female participants between 20 and 35 years of age. Part 4 (age effect) included eight male participants between 68 and 80 years of age. All participants were white and none were of Hispanic or Latino ethnicity.

Safety and tolerability

Lorundrostat was well tolerated and no serious TEAEs occurred during the study. One participant in cohort 3 of part 2 was withdrawn on Day 2 because of an AE of sinus tachycardia (117 bpm) before dosing on Day 2, and he received only one dose of 360 mg lorundrostat on Day 1 (Table 2). The overall incidence of TEAEs was comparable between participants treated with lorundrostat (41/87 [47%]) and those treated with placebo (18/29 [62%]). Across all cohorts, dizziness (of mild intensity) was reported by nine (10.3%) lorundrostat-treated participants compared with one (3.4%) placebo-treated participant. No other trends were identified in the frequency of TEAEs (Table S3) or treatment-related TEAEs (Table 2) across single- or multiple-dose levels of lorundrostat. There were no clinically significant findings with respect to clinical laboratory, vital signs, ECGs, or physical examination.

Pharmacokinetics

Following single oral doses across 5–800 mg, plasma drug concentration–time profiles demonstrated rapid absorption with a median time to maximum concentration (T_{max}) 1.0–1.5 h (Table 3; Figure 1b); the mean terminal half-life ($t_{1/2}$) ranged from 7.9 to 10.5 h, and apparent clearance was comparable, ranging from 18.7 to 27.6 L/h. Over the dose range of 5–800 mg of lorundrostat in part 1, dose proportional increases in systemic exposure in terms of $AUC_{0-\infty}$ were observed as the 95% CIs of the slopes for the AUCs vs. dose included the value 1. In contrast, the estimate of the slope (95% CI) for C_{max} versus dose was 1.104 (1.043–1.164), and a slightly greater than dose proportional increase in C_{max} was observed over the range 5–800 mg (Figure S1). The interindividual variability of exposure parameters (C_{max} and $AUC_{0-\infty}$) were low to moderate (<20%).

In the MAD, after 7 days of once-daily administration of oral lorundrostat doses ranging from 40 to 360 mg, plasma drug concentration–time profiles demonstrated rapid absorption with a median T_{max} 1.5–2.5 (Table 3, Figure 1c). In addition, systemic exposure in terms of the AUC_{0-24} and C_{max} appeared to increase in a slightly more than dose proportional manner over the tested multiple-dose range on Days 1 and 7 (Figure S1). For AUC_{0-24} and C_{max} on Day 1, the estimates of the slopes ranged from 1.143 to 1.146 (95% CI, 1.011–1.281), and for AUC_{0-24} on Day 7, the estimate of the slope was 1.147 (95% CI, 1.060–1.234). A steady state was

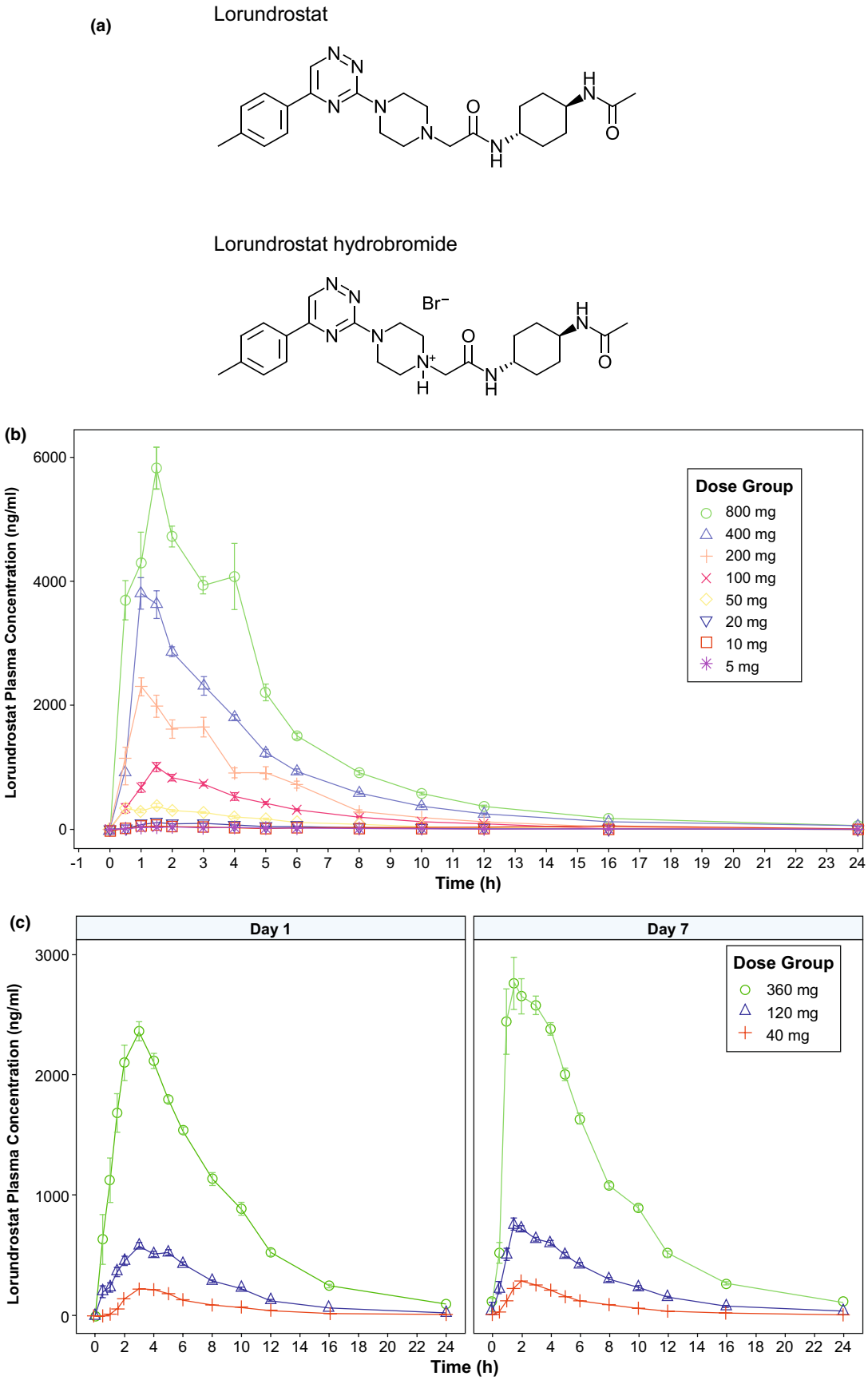

FIGURE 1 (a) Lorundrostat and lorundrostat hydrobromide chemical structures. PK profile (b) SAD administration of lorundrostat by dose group, and (c) MAD administration of lorundrostat by dose group and day. Data shown as mean ± SE. MAD, multiple ascending dose; PK, pharmacokinetic; SAD, single ascending dose; SE, standard error.

TABLE 1 Participant demographics by dose group and phase I study part.

Dose group	Age, years Mean (SD)	Sex, male, *n* (%)	Height, cm Mean (SD)	Weight, kg Mean (SD)	BMI (kg/m^2) Mean (SD)
SAD part 1					
Placebo ($n=16$)	31.2 (12.9)	16 (100)	180.6 (4.7)	81.0 (7.8)	24.8 (2.1)
5 mg ($n=6$)	29.0 (12.3)	6 (100)	178.5 (6.2)	71.5 (5.3)	22.5 (1.8)
10 mg ($n=6$)	31.7 (13.1)	6 (100)	181.2 (6.8)	78.3 (12.4)	23.9 (3.7)
20 mg ($n=6$)	27.5 (12.7)	6 (100)	179.2 (6.4)	76.2 (7.6)	23.8 (2.8)
50 mg ($n=6$)	28.2 (13.8)	6 (100)	184.0 (4.7)	77.3 (7.2)	22.8 (2.2)
100 mg ($n=6$)	27.8 (11.9)	6 (100)	184.0 (3.6)	79.2 (8.7)	23.4 (1.9)
200 mg ($n=6$)	28.3 (11.8)	6 (100)	186.0 (8.1)	81.4 (12.3)	23.4 (2.3)
400 mg ($n=6$)	28.5 (11.9)	6 (100)	187.0 (4.7)	84.2 (6.8)	24.2 (2.8)
800 mg ($n=6$)	33.2 (15.8)	6 (100)	184.2 (7.7)	85.1 (9.9)	25.1 (2.9)
MAD part 2		Sex, male			
Placebo ($n=9$)	29.3 (11.9)	9 (100)	182.4 (7.0)	78.5 (10.2)	23.6 (2.9)
40 mg ($n=9$)	29.2 (10.4)	9 (100)	182.7 (5.4)	76.7 (8.6)	23.0 (2.3)
120 mg ($n=9$)	26.3 (5.6)	9 (100)	184.3 (5.1)	77.8 (12.2)	22.8 (2.6)
360 mg ($n=9$)	30.8 (11.6)	9 (100)	182.6 (7.7)	76.4 (11.7)	23.1 (4.1)
Sex effect part 3		Sex, female			
Placebo ($n=2$)	23.0 (4.2)	2 (100)	164.0 (0.0)	59.9 (8.9)	22.3 (3.3)
100 mg ($n=6$)	25.5 (5.8)	6 (100)	165.5 (4.7)	60.8 (4.7)	22.2 (1.7)
Age effect part 4		Sex, male			
Placebo ($n=2$)	72.5 (3.5)	2 (100)	166.0 (8.5)	77.2 (10.6)	27.9 (1.0)
100 mg ($n=6$)	72.2 (4.1)	6 (100)	176.5 (4.9)	77.9 (6.3)	25.1 (1.9)

Abbreviations: BMI, body mass index; MAD, multiple ascending dose; SAD, single ascending dose; SD, standard deviation.

achieved by approximately Day 5 and very slight accumulation of lorundrostat was observed following multiple dosing compared with Day 1, with the mean rate of accumulation values for AUC_{0-24} ranging from 1.15 to 1.19 across the entire dose range. Consistent with single-dose administration, the mean $t_{1/2}$ values on Day 7 ranged from 9.1 to 11.9 h.

After a single dose of 40–360 mg of lorundrostat on Day 1, the fraction of the dose excreted in the urine as unchanged was low, with mean values ranging between 1.2%–1.8%. The fraction of the lorundrostat dose excreted remained consistent over the dose range tested. Renal clearance (CL_R) was low and generally similar across dose levels (mean CL_R ranging between 0.3–0.5 L/h) after a single dose on Day 1.

After 7 days of once-daily doses of lorundrostat, the fraction of the dose excreted in the urine as unchanged drug was low for all multiple-dose levels, with mean values ranging between 2.3% and 2.6%. The fraction of the lorundrostat dose excreted remained consistent over the multiple-dose range tested. Renal clearance was low and generally similar across all multiple-dose levels (mean CL_R ranging between 0.4 and 0.6 L/h on Day 7).

In part 3 of the study, after a single 100-mg dose of lorundrostat, there was no statistically significant difference between lorundrostat exposure in females and males (Table S4). The median T_{max} values were comparable for female participants (1.3 h) and male participants (1.5 h), and the mean $t_{1/2}$ value was slightly longer for females (12.4 h) than for males (10.0 h).

In part 4, there was no statistically significant difference between older adult and younger adult lorundrostat exposures (Table S4). The median T_{max} value was earlier for older adult participants (1.0 h) vs. younger adult participants (1.5 h), and the mean $t_{1/2}$ value was slightly longer for older adult (11.4 h) than for younger adult (10.0 h) participants.

Based on the dose escalation data of the first cohort in part 2 where lorundrostat was administered in the fed state, the PK profile did not appear to be affected compared with part 1 where lorundrostat was administered in the fasted state.

Pharmacodynamics

All single-dose levels of lorundrostat showed a reduction in PAC compared with baseline at 4 h and 8 h post-dose (Figure 2a; Figure S2). At the lower dose levels (lorundrostat 5–50 mg), the PAC returned to values near

TABLE 2 Treatment-related TEAEs by preferred term.

	Placebo (*n* = 16)	5 mg (*n* = 6)	10 mg (*n* = 6)	20 mg (*n* = 6)	50 mg (*n* = 6)	100 mg (*n* = 6)	200 mg (*n* = 6)	400 mg (*n* = 6)	800 mg (*n* = 6)
SAD Part 1	***n* (%) e**	***n* (%) e**	***n* (%) e**	***n* (%) e**	***n* (%) e**	***n* (%) e**	***n* (%) e**	***n* (%) e**	***n* (%) e**
Headache	2 (12.5) 2	0 (0.0) 0	0 (0.0) 0	1 (16.7) 1	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	1 (16.7) 1	0 (0.0) 0
Somnolence	1 (6.3) 1	0 (0.0) 0	0 (0.0) 0	1 (16.7) 1	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0
Dizziness	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	1 (16.7) 1	0 (0.0) 0
Diarrhea	0 (0.0) 0	0 (0.0) 0	1 (16.7) 1	0 (0.0) 0	0 (0.0) 0	1 (16.7) 1	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0
Nausea	1 (6.3) 1	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0
Asthenia	1 (6.3) 1	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0
MAD Part 2	**Placebo (*n* = 9) *n* (%) e**	**40 mg (*n* = 9) *n* (%) e**	**120 mg (*n* = 9) *n* (%) e**	**360 mg (*n* = 9) *n* (%) e**					
Headache	2 (22.2) 2	1 (11.1) 1	1 (11.1) 1	0 (0.0) 0					
Dizziness postural	0 (0.0) 0	1 (11.1) 1	1 (11.1) 1	1 (11.1) 1					
Somnolence	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	1 (11.1) 1					
Dizziness	0 (0.0) 0	0 (0.0) 0	1 (11.1) 1	0 (0.0) 0					
Dysgeusia	0 (0.0) 0	1 (11.1) 1	0 (0.0) 0	0 (0.0) 0					
Sinus tachycardia[a]	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0	1 (11.1) 1					
Dyspnea	1 (11.1) 1	0 (0.0) 0	0 (0.0) 0	0 (0.0) 0					
Nausea	0 (0.0) 0	1 (11.1) 1	0 (0.0) 0	0 (0.0) 0					
Age effect (≥65 years old) Part 4	**Placebo (*n* = 2) *n* (%) e**	**100 mg (*n* = 6) *n* (%) e**							
Headache	0 (0.0) 0	1 (16.7) 1							

Note: Safety analyses used the safety analysis set (all participants who received at least one dose of study drug). Related TEAEs defined as events with a reasonable possibility to be related to the study drug. There were no treatment-related TEAEs in the Sex Effect Part 3 cohort.

Abbreviations: e, number of occurrences of the treatment-related adverse event; MAD, multiple ascending dose; SAD, single ascending dose; TEAE, treatment-emergent adverse events.

[a]Telemetry monitoring showed a normal heart rate between 60 and 80 bmp during the night and most of the day, with quick increases to 110–130 bpm as soon as vital signs or ECG procedures began. The baseline values for heart rate on the participant's ECGs were relatively high with values of 84, 91, and 105 bpm at screening, Day −3 and followup, respectively. These observations suggest that a stimulus–response reaction to some of the study procedures was more plausible than a relationship to the study drug.

baseline within 24 h post-dose. At the higher dose levels (lorundrostat 100–800 mg), the reduction of PAC was sustained until 24 h post-dose. PAC showed maximum decreases at 4 h and 8 h post-dose of ~80% compared with placebo. Single doses of lorundrostat, ranging from 10 to 800 mg reduced the AUC_{0-24} for PAC in a dose-dependent manner, with statistically significant decreases ranging from 36% to 77% vs. placebo (Figure 2b; Table S5) without effecting cortisol (Figure 2c). In parts 3 and 4, the change from baseline of PAC and serum cortisol concentrations at 24 h after a single 100-mg dose of lorundrostat was similar in male and female participants, as well as in those aged 18–55 years and aged ≥65 years.

Consistent with the SAD cohorts, multiple-dose lorundrostat data demonstrated that for all dose levels tested, PAC decreased in an apparent dose-dependent manner between 2 and 12 h post-dose on Day 7 (Figure 2d). From 24 h after the last dose on Day 7, a clear rebound in PAC ranging from 73% to 163% compared with placebo was observed at all multiple-dose levels of lorundrostat. These increases were sustained at least until Day 10, 3 days after the last dose.

During non-ACTH challenge conditions, no relevant changes from baseline were observed for serum cortisol concentrations and plasma 11-deoxycortisol concentrations during multiple dosing of lorundrostat. At all multiple-dose levels, lorundrostat completely blunted the ACTH-stimulated aldosterone response on Day 6 compared with placebo, while there was no effect on the cortisol response.

Once-daily administration of lorundrostat at doses ranging from 40 to 360 mg increased plasma renin activity and serum 11-DOC compared with placebo (Figure 3a,b). Elevations in serum potassium concentrations due to lorundrostat-mediated reductions in PAC were mild and seen across all doses tested (Figure 3c). These effects

TABLE 3 Summary statistics for plasma PK parameters.

SAD dose group	$AUC_{0-\infty}$, ng × h/mL	C_{max}, ng/mL	$t_{1/2}$, h	T_{max}, h
5 mg	232 (25)	36.7 (6.7)	8.3 (0.7)	1.5 (1.0–3.0)
10 mg	452 (46)	76.3 (20.4)	7.9 (0.9)	1.3 (1.0–1.5)
20 mg	770 (183)	140.6 (76.1)	10.0 (1.9)	1.5 (1.0–3.0)
50 mg	2157 (259)	572.8 (205.7)	9.1 (1.9)	1.0 (0.5–3.0)
100 mg	5376 (495)	1211.0 (248.3)	10.0 (0.9)	1.5 (0.5–3.0)
200 mg	10,660 (3483)	2847.5 (1015.2)	9.9 (3.6)	1.3 (1.0–3.0)
400 mg	17,275 (2906)	4454.8 (862.4)	9.7 (2.3)	1.3 (0.5–3.0)
800 mg	29,385 (3623)	7708.5 (1617.1)	10.5 (2.1)	1.5 (0.5–4.0)
MAD dose group	**AUC_{0-24}, ng×h/mL**	**C_{max}, ng/mL**	**$t_{1/2}$, h**	**T_{max}, h**
40 mg, Day 1	1574.0 (282.0)	252.3 (64.4)	5.3 (0.9)	3.0 (2.0–5.0)
40 mg, Day 7	1795.0 (312.0)	365.1 (46.5)	9.1 (1.8)	2.0 (1.0–3.0)
120 mg, Day 1	4876.0 (1024.0)	886.7 (291.1)	4.4 (0.8)	3.0 (0.5–5.0)
120 mg, Day 7	5816.0 (1315.0)	1038.7 (342.8)	11.9 (2.5)	1.5 (0.5–5.0)
360 mg, Day 1	19335.0 (3111.0)	3220.0 (1171.0)	4.4 (0.5)	3.0 (0.5–8.0)
360 mg, Day 7	21825.0 (3955.0)	3812.1 (1282.3)	9.2 (2.5)	2.5 (1.0–4.0)

Note: All PK analyses used the PK analysis set (all participants who received at least one dose of study drug other than placebo and had PK data considered sufficient and interpretable). Data are shown as mean (SD) except for T_{max} shown as median (minimum, maximum).

Abbreviations: $AUC_{0-\infty}$, area under the curve from 0 to infinity; AUC_{0-24}, area under the curve from 0 to 24 h; C_{max}, maximum concentration; MAD, multiple ascending dose; PK, pharmacokinetic; SD, standard deviation; $t_{1/2}$, half-life; SAD, single ascending dose; T_{max}, time to reach the maximum concentration.

declined during the washout period. Urine sodium and urinary $\log_{10}$ ($10 \times Na^+/K^+$) ratio were increased at all dose levels tested in the SAD, consistent with reduced aldosterone signaling (Figure 3d). In the MAD, the urine sodium and urinary $\log_{10}$ ($10 \times Na^+/K^+$) ratio was initially increased on Day 1 at all dose levels, but the ratio returned to baseline levels at the next sampling timepoint at the end of the dosing period on Day 7. From Day 7 to Day 9, clear reductions in the urinary $\log_{10}$ ($10 \times Na^+/K^+$) ratio were observed, suggesting a rebound effect for the sodium/potassium ratio as well.

DISCUSSION

Lorundrostat is a novel investigational small-molecule inhibitor of aldosterone synthase. Oral administration of lorundrostat, either as single dose up to 800 mg under fasted conditions (SAD part 1) or as once-daily dose up to 360 mg for 7 days under fed conditions (MAD part 2), was well tolerated with a favorable safety profile in a group of healthy male participants. Administration of a single oral dose of lorundrostat 100 mg under fasted conditions was also well tolerated by healthy female participants and healthy older adult male participants (aged ≥65 years).

PK findings from the SAD and MAD studies demonstrated no relevant effects of male vs female sex, older adult vs. younger adult age, or fed vs. fasting state on the PK profile of lorundrostat. Separate from the study we report in this article, a formal food effect study has been conducted and confirms lack of effect of food on the absorption and PK of lorundrostat (tablet formulation; data on file). After single- and multiple-dose lorundrostat administration, plasma levels peaked 1–3 h after administration with a $t_{1/2}$ of 10–12 h. Exposure parameters were generally dose proportional (i.e., $AUC_{0-\infty}$, AUC_{0-24}, C_{max}).

Statistically significant reductions in PAC AUC_{0-24} were observed after administration of single doses of lorundrostat compared with placebo. Reductions were observed from 2 h through at least 12 h post-dose and sustained in a dose-dependent manner up to 24 h at the higher dose levels (≥100 mg). PAC (AUC_{0-24}) was reduced by up to 40% with single 100- to 200-mg doses and up to 70% with single 400- to 800-mg doses. The suppression of aldosterone in all active doses showed an increase at the end of the dosing interval to values that were observed in the pre-dose data for both SAD and MAD, up to 150 ng/dL. A similar pattern of response is expected in patients treated with lorundrostat and is being examined within clinical trials. In contrast to significant reductions in aldosterone, no meaningful effect on serum cortisol levels was observed with any of the doses tested. The lack of effect of lorundrostat on serum cortisol levels was also seen in the MAD part of the study (part 2), for both basal cortisol levels and in response to ACTH stimulation. These findings are consistent with lorundrostat's highly

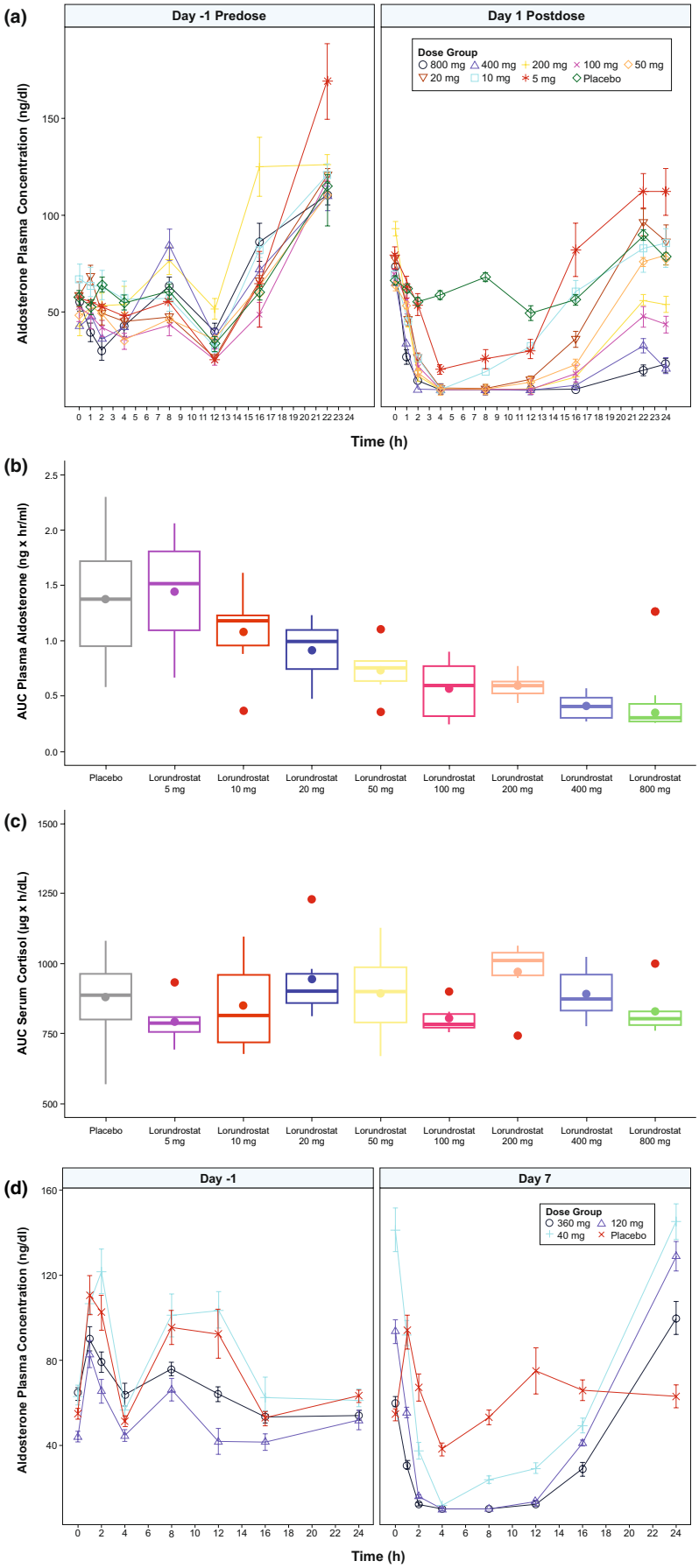

FIGURE 2 (a) Aldosterone time profile for SAD administration of lorundrostat by dose group (Day -1 is the day prior to dosing, reflecting normal circadian rhythm and Day 1 is the day of dosing, showing suppression of plasma aldosterone; mean ± SE). Part 1 SAD (b) aldosterone AUC_{0-24} and (c) cortisol AUC_{0-72} by dose group. (d) MAD administration of lorundrostat by dose group (Day −1 is the day prior to initiating dosing, reflecting normal circadian rhythm and Day 7 is the final day of dosing, showing suppression of plasma aldosterone up to 16 h post-dose; mean ± SE). All PD analyses used the PD analysis set (all participants who received at least one dose of study drug, had at least one post-dose PD assessment, and PD data considered sufficient and interpretable). In Panels (b) and (c), dots within the box plots indicate mean, horizontal lines indicate median, whiskers indicate minimum and maximum values, and dots outside the box plots indicate outliers. Panel (d) Day -1 represents PAC over time in the MAD portion of the study over a 24-h period. Similar to what was observed in the SAD, the normal circadian rhythm of aldosterone that ranges from ~40–150 ng/dL is observed. Panel (d) Day 7 shows suppression of aldosterone in all active doses followed by an increase at the end of the dosing interval to values that are observed in the MAD pre-dose data, and also in the SAD portion, up to 150 ng/dL. AUC, area under the curve; AUC_{0-24}, area under the curve from 0 to 24 h; AUC_{0-72}, area under the curve from 0 to 72 h; PD, pharmacodynamic; SAD, single ascending dose; SE, standard error; MAD, multiple ascending dose.

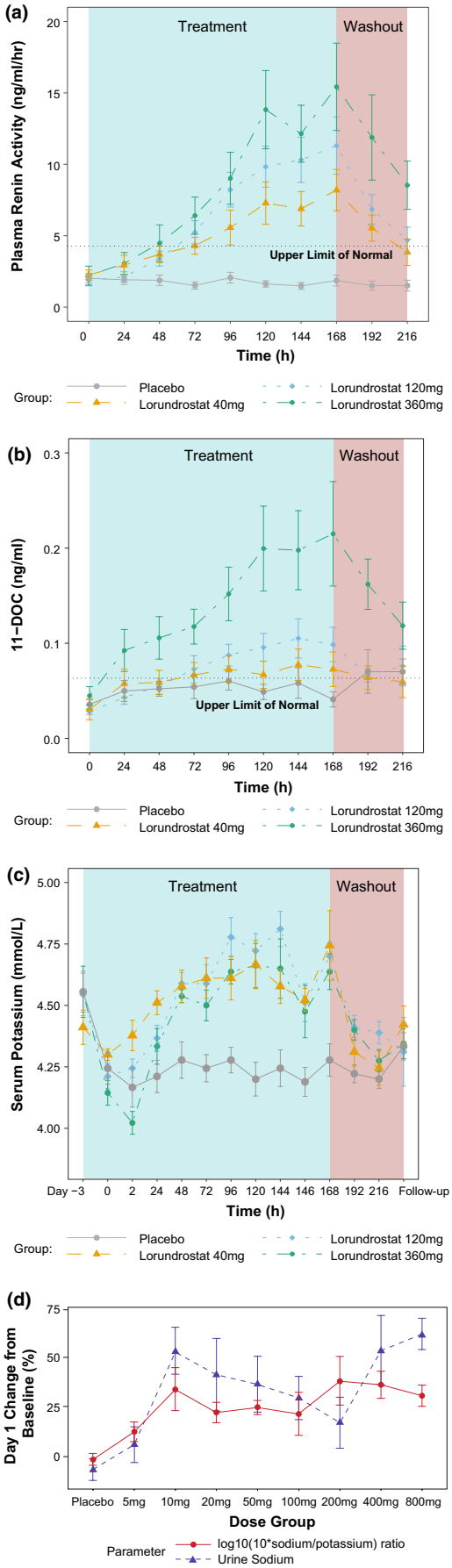

FIGURE 3 Effect of lorundrostat on (a) plasma renin activity, (b) 11-DOC, and (c) serum potassium in part 2 MAD study. Effect of lorundrostat on (d) renal sodium handling in SAD study. Data shown as mean ± SE. Panel (c) Within 2 days of initiation of lorundrostat treatment, serum potassium rose in all three dose cohorts (part 2) by group mean values ranging from 0.28 to 0.38 mmol/L and remained elevated throughout the remainder of the 7 days of treatment. There was a maximum mean increase of 0.44 mmol/L after 40 mg on Day 8, 0.6 mmol/L after 120 mg on Day 7, and 0.48 mmol/L after 360 mg on Day 6. None of the subjects reached the stopping criterion for potassium of 6.0 mmol/L. MAD, multiple ascending dose; SAD, single ascending dose; SE, standard error.

selective (374-fold) inhibition of CYP11B2 vs. CYP11B1 compared with the non-selective aldosterone synthase inhibitor osilodrostat.

Measurement of the effect of lorundrostat on renal sodium excretion provides a direct measure of the effect of aldosterone on renal tubular function, and thus on the potential to reduce intravascular volume and ameliorate volume-dependent systemic hypertension. In the MAD portion of the study, lorundrostat doses of ≥40 mg produced an increase in urinary $\log_{10}(10 \times Na^+/K^+)$ ratio, confirming that the observed reduction in aldosterone had the anticipated functional effect—at least during the initial suppression of aldosterone production. As expected, longer term measurements of Na^+/K^+ ratio in the multiple-dose study tended to return to baseline once new steady-state volume status was achieved.

Although the urinary electrolyte measurements confirmed that there was a brief period of natriuresis at all doses tested, the duration and stability of the sodium depletion is not addressed by that measurement. If suppression of aldosterone is too brief, there could be compensatory sodium–potassium exchange later in the day, with no sustained effect on intravascular volume and serum electrolyte concentrations. Within 2 days of the initiation of lorundrostat treatment, serum potassium rose modestly in all three dose cohorts (part 2) by group mean values ranging from 0.28–0.38 mmol/L and remained elevated throughout the remainder of the 7 days of treatment. Upon cessation of lorundrostat treatment, serum potassium levels rapidly fell toward pretreatment baseline. While hyperkalemia is a known and expected consequence of the RAAS, only modest elevations in serum potassium were observed that were quickly reversible with cessation of lorundrostat consistent with its half-life 10–12 h. Given that the PD effects on renal electrolyte regulation were near maximum at 120 mg daily and the excessive 11-DOC accumulation observed at dose of 360 mg once daily, the data suggest that the dose selection for the treatment of hypertension should avoid the higher dose level and focus on doses up to 120 mg daily.

These results confirm that the PK, PD, and safety profile of lorundrostat supported the progression to a phase

II dose-range finding and proof-of-concept study in individuals with uncontrolled hypertension, including those with treatment-resistant hypertension.[16] The lorundrostat capsule formulation used in the first-in-human studies reported here is equivalent to ~50% of exposure seen in the tablet formulation used in the phase II study (doses used in the phase II study included 12.5 mg once daily and twice daily, 25 mg twice daily, 50 mg once daily, and 100 mg once daily). There is high unmet medical need in this population, with ~77% of the overall US adult population with hypertension failing to achieve the targeted degree of blood pressure control.[17]

AUTHOR CONTRIBUTIONS

M.A.T, S.D., and D.M.R. wrote the manuscript; H.S., Y.O., K.O., S.M.A.R., A.F., Y.H., M.K., K.S-K., M.T.V.I., J.J.V.L., BT.S. and D.M.R. designed the research; H.S, Y.O., K.O., S.M.A.R., A.F., Y.H., M.K., K.S-K., S.M.A.R., M.T.V.I., J.J.V.L., BT.S., and D.M.R. performed the research; H.S., Y.O, K.O., S.M.A.R., A.F., Y.H., M.K., KS-K, M.T.V.I., J.J.V.L., B.T.S., and D.M.R. analyzed the data.

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CONFLICT OF INTEREST STATEMENT

Hidetoshi Shimizu, Yoshiyasu Ohta, Kei Ogawa, Sheikh Mohammed Ashfaq Rahman, Aya Fujii, Yuki Hiraga, Mizue Kawai, and Kanami Sugimoto-Kawabata are employees of Mitsubishi Tanabe Pharma Corporation, which invented and out-licensed the compound to Mineralys Therapeutics, Inc. Thijs van Iersel and Jan Jaap van Lier are employees of ICON, the CRO which conducted the clinical study. BT Slingsby is Board Chairman of Mineralys Therapeutics, Inc. Michael A. Tortorici, Stephen Djedjos, and David M. Rodman are employees of Mineralys Therapeutics, Inc.

DATA AVAILABILITY STATEMENT

All data generated or analyzed during this study are included in this published article (and its supplementary information files).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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